Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18AU56]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

EMISSIONS AND NVH CONTROL

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Name major causes of emissions. Define stoichiometric combustions.	**07**
	b.	List the problems caused by the emissions.	**07**
2.	**a.**	State the mechanism of NO_x formations.	**05**
	b.	Discuss the methods to reduce NO_x emissions in exhaust gases.	**09**
3.	**a.**	Briefly discuss the sources of hydrocarbon exhaust emissions.	**07**
	b.	Explain flame quenching phenomenon?	**07**
		OR	
4.	**a.**	State the effect of following parameter in hydrocarbon emissions i. Engine speed ii. Spark timing iii. Exhaust back pressure	**06**
	b.	Explain evaporative emissions and their control?	**08**
5.	**a.**	Name sources of Noise and vibration.	**06**
	b.	Explain harshness and its effect on passengers?	**08**
		OR	
6.	**a.**	List the types of NVH.	**06**
	b.	Discuss the importance of NVH.	**08**
7.	**a.**	Present the relation between sound pressure level and sound intensity level.	**04**
	b.	Show that the distance from the point source doubles, the sound intensity level decreases by 6DB	**10**
		OR	
8.	**a.**	Define and explain sound spectra?	**04**
	b.	Explain subjective and objective assessment of sound.	**10**